

Slack, setup, and hold

Slack turns tags into a pass or fail

Goldens to remember

- Setup: $R-A = 10-3.2 = 6.8$
- Hold lite: $A-0 = 3.2$
- Positive means the check passes
- Keep these numbers handy, the browser challenges and Track A tests use the same instance

Pseudocode

- Slack pseudocode is arithmetic on tags you already have
- Setup slack is required minus arrival
- Hold lite uses required hold zero at the sink and arrival minus that required
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- On the tiny chain with period ten
- Shrink the period to three and setup fails first

Algorithm sketch — try these

```
INPUT: A[], R_setup[], R_hold[] (lite)
OUTPUT: setup_slack, hold_slack, meet?
setup_slack(p)  $\leftarrow$  R_setup[p] - A[p]
hold_slack(p)  $\leftarrow$  A[p] - R_hold[p]
R_hold[sink]  $\leftarrow$  0 in this lite model
meet_setup if setup_slack(sink)  $\geq$  0
meet_hold if hold_slack(sink)  $\geq$  0
GOLDEN: setup(out)=6.8; hold(out)=3.2
```

Setup slack is required minus arrival

SLACK, SETUP, AND HOLD

Setup slack is required minus arrival

Step 1 / 5 · setup-formula · module02-01-slack-setup-hold



Explanation

At the endpoint out, setup slack equals $R - A$. With $R=10$ and $A=3.2$, slack is 6.8—positive means the path meets the single-cycle check.

- $\text{setup_slack} = R - A$
- Positive $\rightarrow$ meets setup
- Golden at out: 6.8

$R(\text{out})=10$
 $A(\text{out})=3.2$
 $\text{setup slack}=6.8$

Setup slack is required minus arrival

Hold slack uses a different required

SLACK, SETUP, AND HOLD

Hold slack uses a different required

Step 2 / 5 · hold-formula · module02-01-slack-setup-hold



Explanation

For this lite hold model, required at out is 0 on the same edge. Hold slack is $A - R_{\text{hold}} = 3.2 - 0 = 3.2$.

- $\text{hold_slack} = A - R_{\text{hold}}$
- Do not reuse setup required
- Golden hold at out: 3.2

```
R_hold(out)=0  
A(out)=3.2  
hold slack=3.2
```

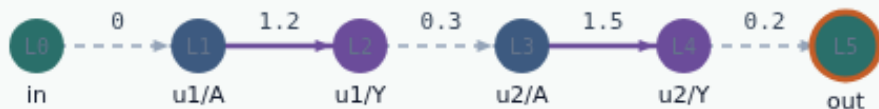
Hold slack uses a different required

The sign tells you pass or fail

SLACK, SETUP, AND HOLD

The sign tells you pass or fail

Step 3 / 5 · sign-matters · module02-01-slack-setup-hold



Explanation

Negative setup slack means the path is too slow for the period. Negative hold means data changes too soon. Read the sign before you chase the path.

- Setup fail → slow path
- Hold fail → fast path / race
- Both can be true on different checks

setup +6.8 (pass)
hold +3.2 (pass)

The sign tells you pass or fail

Report slack at endpoints first

SLACK, SETUP, AND HOLD

Report slack at endpoints first

Step 4 / 5 · endpoint-habit · module02-01-slack-setup-hold



Explanation

Engineers scan worst negative slack at endpoints, then open the path. On this toy netlist there is one sink—out—so both checks live there.

- Endpoints first
- Then open the path
- One sink keeps the lesson clear

```
sink: out  
checks: setup + hold
```

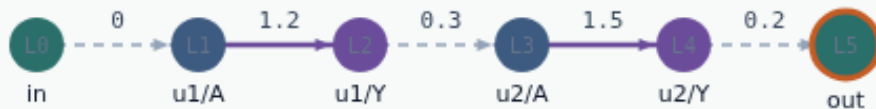
Report slack at endpoints first

Period changes setup, not hold lite

SLACK, SETUP, AND HOLD

Period changes setup, not hold lite

Step 5 / 5 · period-knob · module02-01-slack-setup-hold



Explanation

If the period were tighter, setup required would drop and setup slack would shrink. Hold in this lite model stays tied to arrival versus zero.

- Period feeds setup required
- Hold lite ignores period here
- Full STA hold uses more edges

```
period=10 → R_setup=10  
hold R=0 (lite)
```

Period changes setup, not hold lite

Browser lab track

- In the browser lab, open **slack-setup-hold**
- Load the starter, run the analysis once, and read the metrics panel
- Orient yourself, challenge panel, canvas, Check, then mirror the same goldens in code

Implement track

- In the implement track
- Run ``python3 common/test_propagate.py`` (and the timing-graph test) until the goldens print

Pitfall

- Do not mix setup and hold required maps
- Do not propagate before the graph is levelized
- After an edit or exception, recompute, stale tags lie

Your turn

- Finish the checklist on at least one track, preferably both
- When your numbers match the goldens, take the quiz, then continue